

Docent: Design Considerations for Voice-Led Conversational AI in Cultural Spaces

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Abstract — This project investigates natural conversation as a mode of human-computer interaction with AI systems. The project examines the design considerations to develop conversational interfaces that more closely approximate the responsiveness, coherence, contextual sensitivity, and flexibility of human conversation. To explore these considerations, the project develops Docent, a voice-led conversational AI system to serve as a guide cultural-spaces, where static information delivery is replaced by interactive spoken dialogue. This use-case serves as an experimental vehicle for examining non-linear inquiry, retrieval-grounded explanation, interruption, continuance, conversational memory, and dialogue-driven navigation. The project frames conversation not merely as a convenient input method, but as a complex interactional system involving turn-taking, grounding, shared understanding, and situated action. The resulting prototype and evaluation are used to consolidate design considerations for improving the conversational ability of voice-led AI systems more generally.

Evaluation with four participants achieved 86.4% of the maximum local-conversational-objective score; contextual and pragmatic behaviors performed comparatively strongly, while turn-taking and floor management remained the principal limitations

Keywords — Conversational AI, Voice User Interfaces, Spoken Dialogue Systems, Human-AI Interaction, Human-Computer Interaction, Museum guides

I. INTRODUCTION

Leaps and bounds have been made in the development of increasingly sophisticated Artificial Intelligence systems, with recent models demonstrating significant improvements in reasoning, language generation, and semantic understanding. However, the user interfaces through which people interact with these systems remain largely shaped by earlier text-based and graphical paradigms. Despite the natural language capabilities of contemporary AI models, the primary mode of interaction with these systems is still often a text box, supported by menus, buttons, and screen-based navigation. This creates a gap between the apparent intelligence of AI systems and the relatively constrained methods by which users are required to access that intelligence.

The promise of Artificial Intelligence is not only that machines may become more capable, but that they may reduce the need for users to adapt themselves to machine-centred modes of operation. From the perspective of Human-Computer Interaction, this suggests that progress in AI should not be measured solely by model capability, but also by the extent to which interaction with these systems can fit more naturally into human communicative behaviour.

Natural conversation is therefore a compelling interface paradigm, since it represents one of the most fundamental forms of human communication. However, conversation is not simply the exchange of text through speech. It is a coordinated, situated, and socially managed activity.

This project investigates the design considerations for AI systems that can interact through natural conversation. To explore these considerations in practice, the project develops **Docent**, a voice-led conversational AI system implemented in a cultural-space (e.g. an art gallery or museum). Cultural spaces provide are useful experimental settings because they highlight the difference between static information delivery and interactive human conversation: a visitor in a gallery may ask follow-up questions, interrupt an explanation, refer to objects in the surrounding environment, or move non-linearly between topics. The system therefore serves as a proof-of-concept for examining how conversational responsiveness, coherence, contextual grounding, retrieval-supported explanation, interruptibility, continuance, and dialogue-driven navigation can be supported in voice-led AI systems more generally.

II. BACKGROUND AND RELATED WORK

The background to this project is necessarily interdisciplinary because the problem of conversational AI does not belong to computer science alone. If the aim is to design AI systems that interact through natural conversation, then the relevant literature must include not only technical work on dialogue systems but also work on how human conversation itself is organized. For this reason, this section draws from conversation analysis and sociology, philosophy of language and pragmatics, psycholinguistics and cognitive science, HCI and situated action, in addition to more typical domains expected in a project of the nature such as AI research on spoken dialogue systems, conversational agents, and retrieval-grounded generation. This range of fields reflects the central claim of the project: conversation is not simply a language-generation problem, but an interactional, social, contextual, and technical design problem.

A. Conversation as an Organised Interactional System

Sacks, Schegloff and Jefferson (1974) describe conversation as an organized turn-taking system structured around turn-constructional units, transition-relevance places and locally managed speaker selection. Their account is particularly relevant to spoken AI because determining when to respond cannot be reduced to detecting silence: pauses may occur within unfinished turns, while interruptions and acknowledgements may overlap with another speaker's contribution. This motivates the prototype's separation of acoustic speech detection from turn-completion prediction

and its explicit treatment of floor management and interruption. Their distinction between the context-free and context-sensitive organization of conversation additionally informs the separation between the domain-independent Conversation Core and the domain-specific Docent layer.

B. Grounding and Shared Understanding

Clark and Brennan (1991) describe grounding as the incremental establishment of sufficient mutual understanding for the purposes of an interaction. Consequently, conversational contributions frequently depend upon shared context rather than restating their complete referent. For the present system, this motivates maintaining dialogue history and resolving context-dependent expressions such as “*Who painted it?*” before retrieval and response generation.

C. Conversational Quality and Pragmatics

Grice’s account of conversational implicature provides a useful framework for evaluating the quality of conversational contributions. The cooperative principle and the maxims of quantity, quality, relation and manner describe expectations that contributions should be appropriately informative, sufficiently supported, relevant to the ongoing exchange, and expressed clearly (Grice, 1975). Rather than treating these maxims as deterministic rules to be implemented directly, this project uses them as qualitative evaluative criteria for examining conversational behaviour.

This is particularly useful because conversational quality cannot be adequately represented by a single judgement of whether an interaction was simply “good” or “bad”. Grice’s framework allows individual responses to be examined more specifically: whether the system provided too much or too little information, whether its claims were adequately grounded, whether it remained relevant to the user’s contribution, and whether its response was sufficiently clear for spoken interaction. These criteria consequently inform both the conversational design of the prototype and the qualitative evaluation of its responses.

D. Situated Action and Contextual Interaction

Suchman (1987) argues that action is shaped through the circumstances in which it unfolds rather than simply through the execution of predetermined plans. This is especially relevant to Docent, where visitors move through a physical environment, redirect their attention between artworks and formulate questions in response to what they encounter. The cultural-space implementation therefore avoids assuming a fixed conversational path and supports non-linear inquiry, contextual reference and situated descriptions, while explicitly avoiding claims that the system can itself see or gesture towards the surrounding environment.

E. Retrieval-Augmented Generation and Provenance

Existing spoken-dialogue research demonstrates that turn-taking, dialogue state, repair and grounding can be operationalized computationally, although implementations are frequently tied to particular task or embodiment settings. A further requirement for Docent is factual grounding. Retrieval-Augmented Generation connects language-model responses to external source material rather than relying solely upon parametric knowledge (Lewis et al., 2020). In the cultural-space implementation, this provides curated evidence concerning artworks and artists and allows factual retrieval to be separated from conversational interpretation.

Taken together, this literature identifies several features required for conversational interaction; turn coordination, grounding, pragmatic interpretation, situated behavior and factual grounding, but these concerns are often considered independently or within application-specific dialogue systems. The contribution of this project is to examine how these principles can be brought together within a modular voice-led conversational architecture, while distinguishing between mechanisms that are general to conversational interaction and those that arise from the requirements of a particular domain. The resulting Conversation Core operationalizes the former, while Docent provides a cultural-space implementation through which the latter can be examined. The project additionally evaluates the technical and interactional trade-offs involved in implementing these mechanisms within a real-time speech pipeline

III. SYSTEM DESIGN

Considering again the title “*Docent: design consideration for voice-led conversational AI systems in cultural spaces*” and then drawing on distinction between the context-free and context-sensitive organization of conversation (Sacks, Schegloff and Jefferson, 1974), the system design of the prototype, may understood as embodying two corresponding sets of considerations: “*design consideration for voice -led conversation AI systems (fundamental & general)*” and “*design considerations for voice-led conversational AI systems in cultural space (domain-specific)*”.

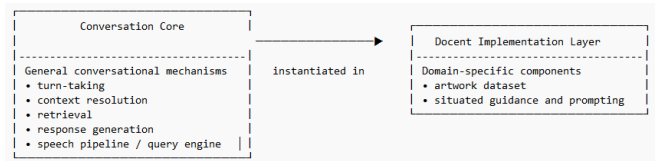


Figure 1. Dual Conversational System Architecture

The architecture of the prototype mirrors this distinction with a general **Conversation Core** implements the mechanisms intended to support conversational interaction independently of any specific domain, while a separate **Docent implementation layer** provides the knowledge and behaviors specific to the cultural-space application it represents the context-sensitive implementation of the more general conversational framework.

A. Software Architecture

The prototype follows a client-server architecture consisting of a **React frontend** and a **Python Fast API backend**. The frontend provides the user-facing interface and manages microphone capture and audio playback, while the backend contains the conversational processing architecture, including speech recognition, turn-completion detection, conversational context resolution, domain retrieval, language-model response generation, and text-to-speech synthesis.

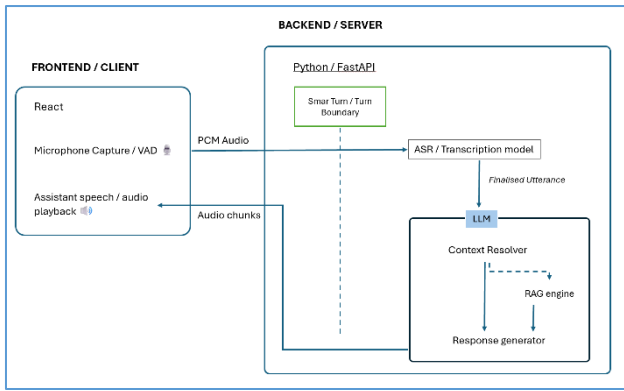


Figure 2. System Overview

For voice interaction, communication between the frontend and backend is principally maintained through persistent WebSocket connections, allowing audio, model responses and synthesised speech to be progressively transmitted during an interaction. The use of this streaming architecture is discussed further in Section V in relation to latency and the technical experiments conducted during development.

B. End-to-End Request Flow

A secondary objective of this project was to determine whether a real-time conversational voice interface could be constructed predominantly from open-source and independently deployable AI components rather than a proprietary end-to-end speech API.

The prototype therefore uses a modular pipeline in which turn detection, speech recognition, conversational processing, language generation and speech synthesis remain separable, allowing individual components to be replaced and experimentally compared during development

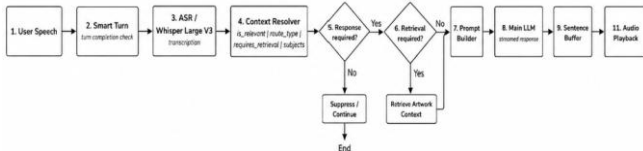


Figure 3. User utterance / request flow

Following a candidate end of speech, accumulated PCM audio is evaluated by **Smart Turn**, an ONNX-based turn-completion model operating directly on audio. Smart Turn determines whether the turn should be finalised while audio is concurrently streamed to the speech-recognition service. The final system uses **Whisper Large V3**, hosted on QMUL's Jupyter infrastructure and accessed through a persistent WebSocket connection; once turn completion is confirmed, the transcription stream is finalised. The transcription layer remains abstracted so that alternative ASRs can be substituted without altering the wider pipeline.

The recognised utterance and recent dialogue history are then passed to the **context_resolver**, which returns *is relevant*, *route type*, *requires retrieval*, and *subjects*. These values determine whether the utterance should affect the conversation, its conversational function, whether external knowledge is required, and the subject inferred from context. Reference resolution therefore precedes retrieval, allowing

context-dependent utterances such as “*Who painted it?*” to retrieve directly against the resolved subject where necessary.

For responses requiring generation, the original utterance, dialogue history, context-resolution output and any retrieved evidence are combined with the Docent behavioural prompt and supplied to the principal language model through Ollama. The response is streamed incrementally, with completed sentences passed to **Kyutai Pocket TTS** while subsequent text continues to generate; synthesised PCM audio is then streamed back to the client for playback. This overlapping pipeline reduces the amount of completed processing required before the user begins hearing a response.

IV. IMPLEMENTED DESIGN CONSIDERATIONS

Following the distinction established in Section III, the design considerations implemented in the prototype are discussed at two levels:

A. Conversation Core: General Design Considerations

The Conversation Core provides the implementation layer through which the general conversational principles identified during the literature review are translated into system behavior. Rather than describing its constituent software components independently, the following subsections consider the interactional problems those components were introduced to address and the way each principle is approximated within the prototype.

It provides a reusable, domain-independent layer for coordinating dialogue turns, conversation state, query handling, retrieval interfaces, and interaction flow. Rather than encoding museum-specific behaviour directly into the core, domain implementations supply their own knowledge, prompts, retrieval services, and actions through defined interfaces.

1. Turn Taking & Transition-Relevance Prediction

A fundamental problem in spoken conversational interaction is determining when one speaker has finished and another may appropriately take the floor. As described by (Sacks, Schegloff and Jefferson, 1974), speaker transition is organised around possible completion points, or transition-relevance places (TRPs), rather than simply around periods in which no sound is being produced. A pause may occur within an unfinished utterance, while a grammatically or pragmatically complete turn may require very little silence before a response becomes appropriate. Consequently, voice-activity detection alone provides insufficient information for deciding when an AI system should respond.

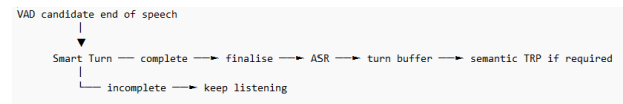


Figure 4. Smart Turn / Turn taking logic

The prototype therefore separates the detection of a pause from the determination that a conversational turn has ended. Voice-activity detection identifies the cessation of speech and establishes a candidate turn boundary, after which the accumulated audio is evaluated by Smart Turn, an audio-based turn-completion model. Smart Turn estimates whether the utterance has reached a point at

which speaker transition is appropriate; where continuation is projected, the system remains receptive to further speech rather than immediately forwarding the utterance for response generation. In this way, turn completion is treated as a problem involving the structure of the user's contribution rather than solely an acoustic threshold.

2. Relevance, Floor Management and Interruption

Detecting that speech has occurred does not establish that the conversational floor should be transferred to the user. In a real environment, the system may encounter background speech or other audio activity that does not constitute a meaningful contribution to the dialogue. Treating every detected utterance as requiring attention would make the system excessively reactive, particularly while the assistant is speaking, causing frequent and unnatural disruption to the interaction.

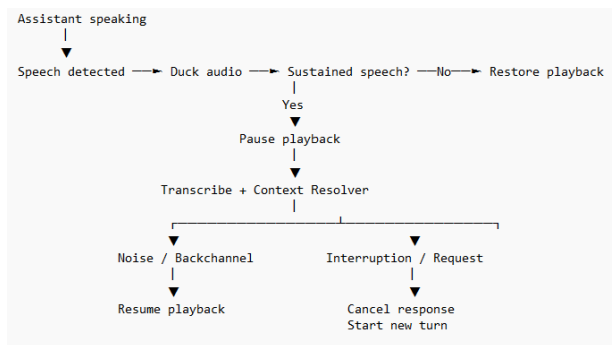


Figure 5. Context resolver process speech and identifies floor grabbing utterances

The Context Resolver therefore provides both an *is relevant* judgement and a *route type*, allowing the conversational system to distinguish speech that should affect the interaction from input that should be ignored. In particular, the *noise route* allows recognized but non-conversational input to be rejected before response generation, while an *interruption* identifies speech intended to displace the assistant's current turn.

Initial user speech or sound will produce a provisional reaction; reducing and/or pausing assistant speech while the utterance is processed. If the resulting utterance is judged to constitute an *interruption*, the existing assistant response is cancelled; if it is judged irrelevant, the response may continue.

3. Grounding Reference Resolution and Shared Understanding

Conversation relies on shared context, meaning that contributions such as “Who painted it?” may be intelligible even when their subject is not explicitly stated. In the prototype, the Context Resolver evaluates each utterance alongside recent dialogue history and extracts the subject or subjects to which the user is referring. If the preceding discussion concerns The Swing, for example, the system can resolve “Who painted it?” to that artwork and retain the resolved subject as part of the conversational state.

This resolved subject can then be passed directly to retrieval system (employing Retrieval-Augmented Generation) where external information is required,

avoiding the need to first rewrite the user's utterance into a self-contained query. Multiple subjects can likewise be resolved and retrieved independently before their evidence is merged. Grounding is therefore implemented primarily through appropriate continuation: where reference can be resolved confidently from shared dialogue context, the system proceeds without requiring explicit confirmation, reserving repair for cases in which sufficient common ground cannot be established.

4. Implicature, Speech Acts and Conversation Function

The conversational function of an utterance is not always equivalent to its literal wording. To account for this, the *Context Resolver* assigns a *route type* that represents what the utterance is doing within the interaction, rather than treating every relevant utterance as a request for a new response. This is particularly important for backchannels such as “yeah”, “right” or “I see”, which may acknowledge the assistant's contribution without seeking to take the floor. Such utterances can therefore be marked as relevant while receiving the *backchannel route type*, causing the Conversation Core to suppress a new generated response and allow the existing turn to continue.

Other route types, including *call_to_action* and *response_request*, allow similarly brief utterances to produce different interactional outcomes according to their inferred function. Pragmatic behaviour is further supported through the *response prompt*, which encourages minimal handling of incomplete contributions and adaptation when a visitor indicates that an explanation was unclear, mistaken or unhelpful. Together, these mechanisms aim to make the system respond to the conversational action performed by an utterance, rather than solely to its literal semantic content.

B. Docent: Domain-Specific Design Considerations

The Docent implementation layer adapts the general Conversation Core to the art gallery use case. While the core provides domain-independent conversational mechanisms such as turn management, context resolution and retrieval, Docent supplies the knowledge and behavioral constraints required for an AI system acting as a gallery guide. The following considerations therefore concern requirements arising specifically from this use case: *how the system should communicate with a visitor situated in front of an artwork, and how domain knowledge should be supplied and grounded.*

1. Situated Interaction and Interpretive Information Delivery

The gallery setting requires Docent to communicate as though addressing a visitor who is physically situated near an artwork, without falsely implying that the system itself can see, point or share the visitor's visual perspective. This behavior is implemented principally through the *Docent system-prompt*, which permits evidence-supported positional language such as “on the left” or “in the background”, while explicitly constraining the model from claiming physical perception or presence. In this way, the conversational behavior is adapted to the visitor's situation without overstating the system's embodied capabilities.

The prompt also defines how information should be delivered in the role of a gallery guide. In keeping with **Grice’s maxim of quantity**, Docent is instructed to provide the main answer first, prioritize a small number of salient interpretive points, and allow further detail to emerge through subsequent questions rather than delivering an exhaustive explanation at once. This constraint was particularly important because, during development, unconstrained LLM responses tended towards greater verbosity than was desirable for spoken interaction. In the intended voice-led form of the system, the visitor cannot skim, reread or visually navigate a response as they might in a text interface; excessive detail therefore risks increasing cognitive load and reducing information retention. The desired interaction is consequently closer to guided face-to-face conversation, in which information is introduced incrementally, and its depth is shaped by the visitor’s continuing interest.

The effectiveness and necessity of several of these prompt constraints, including situated language and response behavior, are examined later in the interaction-oriented experiments of Section V.

2. **Domain Knowledge, Retrieval and Provenance**
Retrieval itself is treated as a capability of the Conversation Core, since access to external knowledge may be required in any conversation where factual accuracy matters. What is specific to Docent is the knowledge source against which that capability operates. The implementation therefore provides a structured artwork dataset containing information such as title, artist, date, descriptive material and interpretive context, which is transformed into vector representations and stored in a persistent retrieval index for reuse across sessions.

This separation supports **adaptability**: the conversational retrieval mechanism remains general, while a different application can replace the artwork corpus and associated domain configuration without altering the underlying conversational architecture. In Docent, the use of curated cultural material additionally provides a basis for factual provenance, reducing reliance on the model’s parametric knowledge when discussing artworks and their historical context.

V. DEVELOPMENT EXPERIMENTS AND TECHNICAL CONSIDERATIONS

The final architecture described in Section III emerged through iterative technical experimentation with alternative models, request profiles and orchestration strategies. Because mechanisms introduced to improve conversational behavior also added processing to an already latency-sensitive voice pipeline, development sought not only to improve behavioral performance, but to do so while minimizing the computational delay introduced by those mechanisms.

A. Latency and its Associated Considerations

Latency represented the principal technical constraint connecting many otherwise independent implementation decisions. Consequently, several experiments investigated

whether conversational reasoning stages could be accelerated, consolidated or moved earlier in the pipeline without substantially reducing behavioral reliability.

1. Turn Completion: Semantic TRP and Smart Turn

The initial implementation treated transition-relevance prediction as a semantic task: following a candidate pause detected by voice-activity detection, the transcription was supplied to a language model which estimated whether the utterance appeared complete or projected continuation. This reflected the project’s initial emphasis on the linguistic structure of turn completion, consistent with Sacks, Schegloff and Jefferson’s account of transition relevance, but introduced an additional sequential inference stage after transcription. Initial experiments were conducted to discover the configuration to deliver the lowest latency metrics

Table 1. Semantic TRP configs comparison

TRP experiment	Accuracy	Median latency	Main finding
Gemma 4 balanced model test	100% (20/20)	1.2980 s	No false waits or premature finalisations
Gemma 4 streamed structured TRP	100% (20/20)	1.1458 s to valid JSON	188 ms conservative paired saving
Original Gemma TRP request	100% (20/20)	7.1845 s	Demonstrated high cost of original request profile

Smart Turn altered this architecture by estimating completion directly from audio, allowing turn prediction to occur before a completed transcript was available. The trade-off was accuracy.

Table 2. Optimal Sematic TRP vs Smart Turn

Method	Evaluation set	Accuracy	Decision latency	Premature finalisations
Semantic TRP — Gemma 4	20 balanced utterances	100%	1.1458 s streamed valid result	0
Smart Turn v3.2	40 human utterances	85%	243.45 ms complete local path	5/20 incomplete cases

2. Conversation routing and context resolution: removing sequential model calls

The original pipeline used a dedicated *utterance classifier* before the main response model, performing much of the routing work now undertaken by the Context Resolver but as an independent sequential model call. Experiments therefore considered both optimization of this classifier and alternative architectures for consolidating its function.

Classifier optimization experiment

Table 3. Experiment i. optimized but external classifier

Configuration	Accuracy	Classifier-only median
Legacy classifier	20/20	2.9459 s
Optimised structured/streaming classifier	20/20	1.8014 s

Routing architecture experiment

Table 4. Compares A: the optimized classifier from experiment I with 2 new architectures; B main model + self-routing and context resolution; C: main model + context resolution handles routing

Architecture	End-to-end routing metric	Result
A - classifier/tool routing	Time to first speech	4.5502s sequential vs 5.9249s unified
B - main-model self-routing	Time to first speech	3.2880s, but only 8/12 correctness
C - context resolver does self-routing	Time to first speech	3.0214 s, with 15/15 correctness

The strongest result came from *Architecture C*, in which a dedicated Context Resolver first established conversational information and retrieval candidates before the main response model was invoked. The present architecture evolved from this result: routing and contextual interpretation are consolidated into a compact think=False Context Resolver, while the main model is reserved principally for response generation; the experimental self-routing stage was subsequently removed. The principal finding was that information required before response generation should be consolidated into the smallest reliable pre-response inference operation.

3. Final Pipeline Optimization

Once the behavioral viability of the individual mechanisms had been established, optimization shifted from individual model calls to the pipeline. Earlier discrete request-response operations were progressively replaced by persistent and streaming communication: audio is transmitted while the user is speaking, transcription remains active until turn completion, language-model output is streamed as it is generated, and completed sentences can be passed to speech synthesis while subsequent generation continues. The resulting architecture therefore replaces a predominantly sequential *speech* → *transcription* → *reasoning* → *completed response* → *synthesis* → *playback* pattern with overlapping processing, shifting the optimization objective towards *time to first useful conversational output*.

These experiments were assessed through repeated manual interaction rather than equivalent controlled quantitative benchmarks

B. Behavioural Experiments

1. Speech Recognition Quality and Compute Constraints

Speech-recognition development was driven principally by **transcription quality**. Moonshine provided lightweight local streaming on CPU hardware, but repeated use exposed recognition errors sufficient to affect downstream contextual interpretation. Whisper Large V3 was therefore explored as a higher-capacity alternative; because its compute requirements exceeded those available locally, it was deployed through QMUL's Jupyter infrastructure, while the common ASR interface retained Moonshine as a fallback. As a formal paired accuracy comparison was not completed, this change is treated as an implementation observation reflecting a **recognition-quality versus compute trade-off**, rather than evidence that Whisper Large V3 was experimentally proven superior.

2. Prompt Behavior and Prompt Necessity

Prompt experiments examined which conversational behaviors required **explicit instruction** and which emerged adequately from the language model without it. This approach was informed by Sacks, Schegloff and Jefferson's (1974) observation that recurrent organizational features of conversation are discoverable within naturally occurring talk. By extension, it was hypothesized that an LLM trained on large corpora containing human discourse may already encode some of these conversational regularities. Selected behaviors were therefore tested with and without dedicated instructions, with prompt rules retained only where they produced a meaningful improvement.

Table 5. Prompt necessity evaluation

Behaviour	Without dedicated prompt	Effect of prompting	Final treatment
Implicature	Performed reliably	No meaningful improvement	No dedicated prompt retained
Repair / adaptation	Less reliable	Improved repair behaviour, although the model could still over-admit errors it had not made	Prompt retained
Fact vs interpretation / hedging	Less consistent distinction	More consistent use of hedging for interpretive claims	Prompt retained
Situated language	Less appropriate to the gallery setting	Clear improvement in positional and context-sensitive description	Prompt retained
Backchannel handling	More likely to generate unnecessary responses	Better recognition of acknowledgements as signals to continue rather than requests for a new answer	Prompt retained

Digression and resumption	Performed adequately	No meaningful improvement	No dedicated prompt retained
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VI. USER STUDY & EVALUATION

Four participants evaluated Docent in a controlled simulated-gallery environment. The study took place in a booked room at Queen Mary University of London using printed reproductions of the artwork available to the guide. Participants were free to inspect the paintings, move between them, ask questions and otherwise interact with Docent as they would with a human gallery guide. The system interface was deliberately visually deemphasized, with the display brightness reduced so that participants could not rely on captions or dialogue transcripts and interaction remained primarily conversational. This was particularly important during conversational breakdowns: participants responded to what they had actually heard rather than being able to inspect the system transcription and anticipate that they had been misrecognized. A fuller description of the procedure and study conditions is provided in the Study Protocol. Evaluation combined dialogue transcripts, technical telemetry, post-task questionnaire responses and semi structured interviews. It was conducted in three stages: an adapted PARADISE analysis quantified local conversational success, dialogue costs and user satisfaction; a conversational-principles analysis assessed the interaction behaviors targeted by the system design; and finally, observed failures were traced back to their likely technical or architectural origins.

Ethical approval for the study was granted by the QMUL EECS Research Ethics Committee; the approval and study materials are reproduced in Appendix A.

A. Evaluation Methodology

1. PARADISE Evaluation

PARADISE (Paradigm for Dialogue System Evaluation) (Walker, Kamm and Litman, 2000) is a framework for evaluating spoken dialogue agents by modelling performance as a combination of task success and dialogue costs, with user satisfaction providing the external criterion against which these factors are assessed. The framework was developed to permit comparison between dialogue strategies while separating what a user achieves from the dialogue behavior used to achieve it. In its original form, PARADISE uses multivariate regression to estimate the relative contribution of success and cost variables to user satisfaction (Walker et al., 1998). Unlike the bounded information-retrieval tasks considered in the original PARADISE studies, Docent supports exploratory conversation without a single terminal task. Task success was therefore operationalized at the level of the local conversational objective (LCO): the immediate objective that the participant attempted to accomplish through one or more dialogue turns

Local conversational objectives

An LCO began when the participant expressed an information or conversational objective and ended when that objective was satisfied, abandoned, or superseded by another objective. Clarifications and reformulations of the same request were retained within the same LCO rather than counted as

additional objectives. Purely social openings, acknowledgements and closings were excluded

Table 6. Participant's LCOs

Participant	LCOs
P 01	55
P 02	22
P 03	14
P 04	19
Total	110

Each LCO was rated using a three-level task-success scale: 2 = objective successfully satisfied; 1 = objective materially addressed but only partially satisfied; 0 = objective not meaningfully satisfied

Table 7. System Success rate

	Successful	Partial	Failed	Points / Max	LGs
P 01	41	13	1	95/110	0.864
P 02	14	8	0	36/44	0.818
P 03	10	4	0	24/28	0.857
P 04	16	3	0	35/38	0.921
Total	81	28	1	190/220	0.864

The system therefore achieved 86.4% of the maximum local-goal-success score, indicating that most participant objectives were ultimately satisfied

Dialogue costs

PARADISE treats task success separately from the interaction cost incurred in achieving it. Dialogue costs were therefore derived independently from the transcripts and telemetry. These included ASR errors, user repairs, failures of contextual continuity, additional turns required to resolve an objective, response latency, abandoned objectives, interruption or turn-fragmentation events, and audio-buffer underruns. Where the transcript included both an ASR transcription and an Actual utterance, the latter was treated as the spoken ground truth. This allowed ASR failure to be distinguished from semantic success: a recognition error could occur while the system nevertheless inferred the intended request and answered correctly.

	ASR Err. Rate	Repair rate	Context Failure	Turns to resolution	Means response latency
P 01	25.5%	9.1%	0.0%	1.13	2.07s
P 02	9.1%	0.0%	0.0%	1.00	4.33s
P 03	28.6%	28.6%	21.4%	1.43	13.08s
P 04	21.1%	0.0%	0.0%	1.00	7.50s

The separation of ASR error from dialogue success was particularly informative. P-01 and P-04 contained several inaccurate transcriptions that nevertheless resulted in an appropriate response, whereas missed utterances in P-03 propagated into incorrect conversational context and subsequent repair.

User satisfaction

User satisfaction was calculated from the 15 administered Likert items; negatively worded items 15–18 were reverse-coded before averaging. The pre-task questionnaire was used only for participant background and was not incorporated into the evaluation score

	Post-task total	Maximum	Mean satisfaction / 7
P-01	77	105	5.13
P-02	78	105	5.20
P-03	71	105	4.73
P-04	98	105	6.53

The quantitative measures were broadly consistent with the post-task interviews. P-04 produced both the highest Local Goal Success score and the highest user-satisfaction score. By contrast, P-03 achieved a comparable level of task success to P-01 but exhibited considerably greater latency, repair and contextual failure, corresponding with the lowest satisfaction score and explicit complaints regarding response delay and topic transitions.

Normalisation and limitation of the PARADISE model

PARADISE normally standardises performance measures and uses multivariate regression to estimate their contribution to user satisfaction. The measures in this study were similarly normalised to permit descriptive comparison. However, the study contained only four independent participant-level satisfaction observations. Estimation of a regression-derived performance function from this sample would therefore be statistically unreliable and highly susceptible to overfitting.

The evaluation consequently adopts PARADISE as a descriptive evaluation framework rather than attempting to derive generalisable performance weights. This retains its distinction between task success, interaction cost and user satisfaction while avoiding unsupported statistical inference.

A particularly clear result is provided by P-01 and P-03. Their Local Goal Success scores were almost identical (0.864 and 0.857 respectively), yet P-03 exhibited substantially greater repair, contextual failure, turns to resolution and latency and reported lower satisfaction. This demonstrates that successful completion of conversational objectives alone did not adequately characterise the quality of the spoken interaction.

2. Conversation-Principles Evaluation

PARADISE primarily evaluates whether objectives were achieved and at what interaction cost. A second analysis therefore considered whether the dialogues exhibited the conversational behaviors targeted by the design of Docent. These principles were derived from the interaction requirements established earlier in the project: naturalistic turn-taking, floor management, grounding and repair, contextual continuity, pragmatic interpretation, dialogue-driven navigation, and appropriately scoped information delivery.

Each principle was evaluated only when the dialogue provided a genuine opportunity for that behavior to be demonstrated. Opportunities were rated as 2 = successfully handled; 1 = partially handled; 0 = failed. The score for each principle represents the points obtained divided by the maximum available points. Opportunity counts are retained because some behaviors, particularly interruption and floor management, occurred considerably less frequently than ordinary contextual follow-ups or information requests.

Evaluation criteria

Conversational performance was assessed across seven principles derived from the system's design objectives. *Turn-taking* considered whether turn completion was detected

without premature finalization or loss of the user's contribution; *floor management and interruption* considered handling of overlaps, continuations and changes in floor ownership; *grounding and repair* assessed recognition and recovery from misunderstanding, ambiguity or correction; *contextual continuity* assessed maintenance and appropriate updating of artwork, topic and referential context; pragmatic interpretation considered indirect, elliptical or context-dependent meaning; *dialogue-driven navigation* assessed movement between artworks or conversational directions through natural dialogue rather than explicit interface control; and *information delivery* considered whether responses provided an appropriate amount of information without material under- or over-answering.

Performance

Principle	Overall opportunities	Points / Maximum	Overall score
Contextual continuity	65	122 / 130	93.8%
Information delivery	110	199 / 220	90.5%
Pragmatic interpretation	42	74 / 84	88.1%
Dialogue-driven navigation	31	54 / 62	87.1%
Grounding and repair	21	31 / 42	73.8%
Turn-taking	12	14 / 24	58.3%
Floor management / interruption	7	7 / 14	50.0%

The results indicate a distinction between the system's semantic conversational capabilities and the mechanics of real-time spoken interaction. Contextual continuity, pragmatic interpretation, dialogue-driven navigation and information delivery all achieved scores above 85%, suggesting that when a usable utterance reached the conversational pipeline, the system generally maintained context and responded appropriately to the participant's intended direction.

The weakest measures were turn-taking and floor management. These failures were concentrated particularly in P-03, where several participant utterances were cut off or missed completely, including transitions between artworks. This frequently produced a secondary contextual failure: the response generated was coherent relative to the context available to the language model but was no longer relevant to the participant's actual conversational state.

VII. FAILURE ANALYSIS / LIMITATIONS

The dialogue transcripts, telemetry and participant interviews were therefore examined together to identify recurring failure families.

1. ASR misrecognition

Evidence from evaluation. Names, artwork titles and ordinary phrases were sometimes transcribed incorrectly. In several cases, however, contextual information allowed the system to infer the intended artwork or request and respond appropriately despite the transcription error.

Diagnosis. Recognition errors originate at the ASR layer, but the architecture currently provides limited domain

information to recognition. Greater exposure to conversational context and an index of likely artwork titles, artists and named entities could improve recognition of uncommon or domain-specific terms.

2. Complete or partial loss of user speech

Evidence from evaluation. Some utterances appeared to be entirely absent from the effective conversation, particularly during P-03. Telemetry showed that at least some of these utterances had in fact been captured and transcribed before the associated turn was subsequently cancelled.

Diagnosis. The speech was therefore not necessarily lost at capture or transcription. Instead, an already recognised contribution could be discarded after the participant resumed speaking. Although closely related to premature turn finalisation, this was retained as a separate failure because its observable effect was different: from the participant's perspective, Docent appeared simply not to have heard part of the interaction.

3. Premature turn finalisation

Evidence from evaluation. Contributions intended as a single utterance were occasionally split, with incomplete fragments being marked as final before the participant had finished speaking. Telemetry showed examples in which incomplete speech was finalised following only a short pause.

Diagnosis. Turn-completion detection was occasionally too aggressive. Processing could therefore begin during a natural intra-turn pause, after which continued speech was interpreted as a new contribution. The evidence suggests that this behaviour arose from the earlier audio turn-completion pathway rather than the longer 3,000 ms fallback thresholds.

4. Lack of visual and situated grounding

Evidence from evaluation. Participants occasionally asked questions about directly visible details—such as the number of figures in a painting or the identity of a visible object—that were absent from Docent's textual artwork information. The system could therefore discuss the painting extensively while still being unable to answer something immediately apparent to the participant.

Diagnosis. Docent currently represents artworks primarily through textual records rather than through visual representations. In effect, the guide has extensive descriptive knowledge of the works but has not itself “seen” them. A future multimodal retrieval corpus could incorporate image-derived information alongside textual RAG content without necessarily requiring continuous live computer vision.

5. Insufficient guide initiative and knowledge flexibility

Evidence from evaluation. Participants expressed interest in more proactive follow-up, stronger connections between artworks and recommendations informed by earlier interests. The current system could also become overly conservative where retrieved evidence did not explicitly contain the requested information.

Diagnosis. The present architecture prioritises tightly grounded local question answering over broader guide-like

initiative. Its retrieval policy and prompting therefore favour avoiding unsupported claims, but this can restrict thematic continuation, recommendation and flexible use of available knowledge. Richer long-range conversational memory and a more nuanced policy for deciding when retrieved evidence is necessary would support a more active guide role.

VIII. FUTURE WORK

A. Corrective improvements

The most immediate future work concerns limitations exposed during user evaluation. Turn-taking requires further calibration and testing; particularly around natural intra-turn pauses and continuation-projecting speech. More robust completion thresholds and fallback finalization strategies could reduce premature finalization and the downstream loss of speech and conversational context. Behaviorally, Docent also requires a stronger conception of its role as a guide rather than merely a respondent. Future development should refine when the assistant should take conversational initiative, make meaningful thematic connections or recommendations, and invite further inquiry, rather than relying on formulaic follow-up questions.

B. Further Architectural Exploration

Broader research directions concern speech recognition, fine-tuning and multimodality. More robust ASR for noisy and unpredictable environments would benefit the Conversation Core generally, since ordinary conversation frequently occurs outside controlled acoustic conditions. Fine-tuning on conversational and guide-like interaction could reduce dependence on explicit prompting and potentially internalize contextual or routing behaviors currently handled through separate inference stages. Finally, multimodality offers two opportunities: augmenting retrieval with visual representations or image-derived artwork information and exploring native speech-to-speech models that bypass the current transcription-and-synthesis cascade, potentially reducing latency while preserving prosodic information lost through text.

IX. CONCLUSION

This project investigated how voice-led AI systems can support more natural conversational interaction, using Docent as a cultural-space implementation of a broader conversational framework. The resulting architecture distinguishes general conversational mechanisms, implemented through the Conversation Core, from domain-specific behaviors supplied by Docent. Evaluation showed that contextual continuity, pragmatic interpretation, information delivery and dialogue-driven navigation were comparatively strong, while turn-taking and floor management remained the principal weaknesses. The central conclusion is therefore that conversational AI quality depends not only on language-model capability, but on the surrounding interaction architecture governing when the system listens, speaks, maintains context and adapts its behavior to the domain.

X. ACKNOWLEDGEMENT

To Anna: Computer science is, in many respects, an art form: it allows one to imagine creative solutions and then bring them into being. As with every art form, however, the practitioner can sometimes become carried away by the process of exploration and neglect the need to conduct experimentation in a timely and disciplined manner. It is for this reason that I am especially grateful to Anna. She continually kept me on track, enabling me to remain focused on answering the research question and delivering the project, rather than pursuing the many caprices that the work repeatedly inspired in me. Thank you for your guidance, your steady hand, and your invaluable advice.

XI. BIBLIOGRAPHY

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MSC PROJECT - REFLECTIVE ESSAY

Project Title:	Docent: Design considerations for voice led conversational AI systems
Student Name:	Olufemi Davies
Student Number:	250958047
Supervisor Name:	Dr Anna Xambo Sedo
Programme of Study:	MSc. Artificial Intelligence

Introduction

This reflective analysis considers the development and evaluation of the conversational AI system, with particular attention to the decisions that shaped its architecture and the lessons that emerged through implementation. The project sought to translate theoretical principles of human conversation into a modular voice-based system while retaining sufficient flexibility for individual components to be examined, replaced and improved. The resulting architecture demonstrated several advantages, particularly in relation to modularity and technological independence, but also exposed practical limitations surrounding latency, reliability and deployment. The following reflection considers these strengths and weaknesses, identifies possibilities for further research and development, examines work that could have been undertaken with additional time, and evaluates the relationship between the theoretical principles informing the project and the practical system that was ultimately produced. It also considers the wider legal, social, ethical and sustainability implications of the design choices made.

Analysis of Strengths and Weaknesses

One of the principal strengths of the project is the systematic and modular approach taken to its architecture. From the outset, the system was designed around open-source and interchangeable components rather than being tied to a single proprietary technology or fully integrated service. Individual components, including ASR, context resolution, retrieval, language generation and TTS, can therefore be replaced or modified independently. This makes the architecture particularly adaptable, as improved models or approaches can be incorporated without requiring the entire system to be redesigned.

This modularity also enables a more deliberate and granular approach to conversational interaction. Components such as the context resolver, and the previously implemented utterance classifier, could be inserted between stages of the pipeline to perform specific functions before an utterance reached the main language model. This degree of control would be considerably more difficult within a fully integrated speech-to-speech system. It allows conversational behaviour to be investigated not simply as an emergent property of a language model, but as something that can be explicitly supported through the surrounding architecture.

The principal weakness of this approach is latency. Every additional stage creates another opportunity for delay, and a modular pipeline cannot necessarily achieve the responsiveness of a system in which speech recognition, reasoning and speech synthesis have been optimised as a single integrated process. This is particularly significant because responsiveness was itself one of the central concerns of the project. The architecture therefore contains an inherent tension between the additional control provided by modularity and the interactional cost created by that control. Components such as context resolution can make the behaviour of the system more explicit and controllable, while simultaneously increasing the amount of processing required before a response can begin. A related weakness is the reliability of response timing. Although substantial reductions in latency were achieved during development, response times are not yet sufficiently consistent for dependable real-world deployment. Occasional delays can be particularly disruptive within spoken conversation, where the appropriateness of a response depends not only on what is said but also on when it is delivered. Consequently, improving reliability remains as important as reducing average latency.

Presentation of Possibilities for Further Work

One important direction for further research concerns the performance of automatic speech recognition under realistic acoustic conditions. The evaluation was conducted under comparatively controlled conditions, whereas an actual gallery environment may contain simultaneous conversations, footsteps, reverberation and other forms of background noise. ASR performance observed during the present evaluation is therefore unlikely to represent its performance during real-world deployment.

Further research could systematically compare different ASR systems under representative gallery noise conditions and examine how transcription errors propagate through subsequent stages of the conversational

pipeline. An incorrectly recognised utterance can affect context resolution, retrieval and ultimately the generated response, meaning that errors introduced at the beginning of the pipeline may influence the entire interaction. Investigating this relationship would provide a more realistic assessment of whether the architecture remains effective within its intended environment.

Further work could also explore richer forms of retrieval. The current retrieval system is predominantly semantic and textual despite operating within a highly visual domain. Image representations could be incorporated into the vector space so that retrieval reflects visual characteristics of artworks alongside their textual descriptions. This would not necessarily require the conversational model itself to become fully multimodal. Instead, visual information could become another retrievable source of contextual information, allowing the system to recognise similarities in composition, objects, colour or other visual characteristics that may not be adequately represented through textual metadata alone.

A further possibility would be the deployment of the system as a hosted web application. This would make the system more accessible beyond the development environment and enable evaluation under more representative patterns of use. Such deployment would also expose additional questions surrounding scalability, reliability and system behaviour across different hardware and network conditions.

Work That Could Have Been Conducted with More Time

Additional development time would have allowed a broader investigation of locally deployable models. The successful deployment of a larger ASR model on QMUL computing infrastructure demonstrated that greater computational resources were available than had initially been practical during development. Earlier stages of the project were restricted by access to cloud computing resources and by the limitations of available local hardware. Further experimentation could therefore have compared additional ASR, language and speech-generation models using QMUL infrastructure, potentially identifying alternatives offering improved accuracy or latency without introducing reliance on external cloud services.

Another area that could have been investigated more extensively is whether explicit context resolution is required within the critical response path. The present architecture performs context resolution before generating a response. An alternative approach would be to rely more heavily upon the language model to resolve conversational context and then make subsequent system behaviour dependent upon structured information contained within its output. This could potentially remove an entire inference stage and therefore reduce latency. Under such an approach, behaviours such as backchannels, noise rejection or interruption handling could potentially be expressed through structured response formatting or tool calling. A recognised backchannel, for example, could result in the model indicating that no spoken response should be generated, allowing the system to continue its existing turn without a separate classification stage. This approach would involve trading some explicit architectural control for a potentially faster pipeline. The available development period did not allow sufficient experimentation to determine whether this reduction in latency would compensate for the corresponding increase in dependence on the behaviour of the language model.

Additional time would also have allowed the pursuit of the project's more ambitious latency target. While considerable progress was made in reducing end-to-end response time, further profiling, model experimentation and architectural simplification would have been necessary to determine how closely the system could approach a response latency of approximately 200 milliseconds.

Critical Analysis of the Relationship Between Theory and Practical Work Produced

The relationship between the theoretical principles informing the project and the practical system produced became particularly apparent through the problem of timing. In theory, the architecture provides mechanisms for managing interruptions, backchannels, context and turn-taking in ways that reflect the conversational principles identified in the literature. In practice, however, these mechanisms depend upon the system being capable of producing the appropriate behaviour at the appropriate moment.

The user evaluation demonstrated several occasions in which an otherwise appropriate system behaviour occurred later than intended because of latency elsewhere in the pipeline. Importantly, these delays were not necessarily produced by language-model inference itself. They could arise from ASR, TTS or another component within the processing chain. A response can therefore be semantically appropriate while still being interactionally inappropriate if it arrives after the conversational moment in which it was required.

This illustrates an important distinction between describing conversational behaviour theoretically and implementing it computationally. Theoretical models of conversation can describe behaviours such as turn-taking, grounding and interruption according to their communicative functions, whereas implementation exposes the

extent to which those functions are dependent upon timing. A backchannel delivered after an excessive delay, for example, may contain suitable content but no longer perform the conversational function of a backchannel. The practical implementation therefore revealed that latency should not be regarded simply as a technical performance metric. Within a voice-based conversational system, latency forms part of the interaction itself. The success of the theoretical architecture consequently depends upon whether its mechanisms can operate within sufficiently narrow temporal constraints. This represents one of the most significant relationships between the theoretical and practical components of the project: implementing theories of natural conversation demonstrated that computational timing is integral to whether those theories can be operationalised successfully.

Awareness of Legal, Social, Ethical and Sustainability Issues

The project also raises legal, social, ethical and sustainability considerations concerning the infrastructure used to construct conversational AI systems. The decision to prioritise open-source technologies made development more difficult in several respects, particularly where proprietary systems might have provided more mature or highly integrated alternatives. Nevertheless, this approach provides benefits in relation to technological independence, transparency, data privacy and AI sovereignty.

A locally deployable architecture reduces dependence upon external service providers and provides greater control over where conversational data is processed. This is particularly relevant for a voice-based system, as recordings and transcriptions may contain personal information even where the intended interaction itself is relatively innocuous. Reducing the need to transmit such information to external services can therefore provide greater control over data handling and privacy.

The modular architecture also reduces dependence upon individual commercial providers. Components can be replaced without redesigning the complete application, mitigating the risk that changes to pricing, availability, licensing conditions or proprietary services render the system unusable. In this respect, the use of open-source technologies contributes not only to technical flexibility but also to the longer-term sustainability and maintainability of the system.

However, technological sovereignty does not remove wider ethical or sustainability concerns. Local deployment still requires computational resources, and the use of increasingly large models creates trade-offs between performance and computational demand. Improvements in accuracy or conversational ability must therefore be considered against their resource requirements. Similarly, increased control over user data creates a responsibility to ensure that locally retained data is itself handled appropriately and securely.

Overall, the project demonstrates both the advantages and limitations of approaching conversational AI as a modular and technologically independent system. Its architecture provided the flexibility and granularity necessary to investigate conversational mechanisms in detail, while its implementation exposed the temporal and computational costs associated with that flexibility. The resulting work therefore provides not only a functional prototype but also a clearer understanding of the engineering, interactional and ethical trade-offs involved in translating theories of human conversation into a practical voice interface.

XII. APPENDIX A

I. Study Protocol Document

Study Protocol Title of study: Evaluating Design Considerations for Voice-Led Conversational AI Guides in Cultural Spaces Context and description This study evaluates Docent, a conversational AI guide designed to deliver an art-gallery or cultural-tour experience through spoken interaction. The system is intended to behave less like a conventional museum app or static audio guide, and more like a responsive guide: answering questions, maintaining context, supporting follow-up discussion, and allowing the user to move through the experience conversationally. Docent implements the answer the wider thesis asks i.e. what design considerations are needed to make interaction with an automated conversational system feel natural and useful in practice. The experiment therefore evaluates how effectively the implemented design choices can deliver the optimal conversational experience of Human-Computer Interaction. Some of these design choices include conversational memory, grounding, turn-taking, interruption handling, continuation, and speech-based navigation of the application. The project is situated within cultural spaces such as museums, galleries, and exhibition environments because these settings provide a useful test case for evaluating usually human guided conversation with an automated system. A human guide in a gallery does not simply deliver fixed information; they respond to questions, adjust explanations, manage attention, and sustain a thread of discussion. This makes the cultural-guide scenario suitable for testing whether the design considerations implemented in Docent produce a natural, coherent, and guide-like interaction. The study will use prepared gallery or exhibition material in a booked room at Queen Mary University of London. This allows the project to simulate a cultural-guide interaction while keeping the study environment consistent for all participants. The purpose of this evaluation is to examine whether the design choices implemented in Docent translate into an experience that feels natural, coherent, responsive, and suitable for cultural guidance; A scenario for which human conversation is considered the gold standard. The interface being evaluated is primarily the quality of the spoken interaction itself: how conversation allows the user to explore content, navigate the system, and receive guidance. Eligibility Participants must: • be over 18; • be able to take part in a spoken interaction in English; • be willing to use an AI guide through speech; • be willing to complete a short questionnaire and interview after the interaction. No specialist background is required. Participants do not need prior knowledge of art, museums, AI, or computer science. Participants whose first language is not English may take part, provided they feel comfortable completing a spoken interaction in English. Because the current prototype is speech-based, participants who are unable to take part in spoken or auditory interaction may not be suitable for this version of the study. This limitation will be acknowledged as an area for future accessibility-focused design work. Planned sample size The study aims to recruit approximately 12–40 participants. 1	This range is appropriate for an exploratory HCI/user-evaluation study of a prototype system, where the aim is to identify patterns in interaction, user experience, breakdowns, and design implications rather than to produce statistically generalisable results. A smaller sample within this range would still allow recurring themes to be identified through questionnaires, transcripts, observation notes, and interviews, while a larger sample would provide broader variation across speaking styles and user responses. Location The study will take place in a booked room at Queen Mary University of London. All participants will complete the study in the same type of controlled environment, using the Docent prototype and the same prepared gallery or exhibition material. Using the same venue for all participants will help keep the study conditions consistent. This reduces the number of uncontrolled factors that could affect the interaction, such as background noise, public visitors, movement through different spaces, venue restrictions, or inconsistent network conditions. Although the project is motivated by cultural spaces such as galleries and museums, the user study will be conducted in a controlled room-based setting so that the evaluation can focus on the participant’s interaction with Docent rather than differences between locations. Setup Participants will use Docent preloaded on a mobile device along with headphones provided by the researcher. Audio recordings may be temporarily retained only to verify transcription accuracy and will be deleted once the transcript has been checked. Developmental testing Standard developmental testing will be carried out during the implementation of the application to ensure that the main technical features operate reliably before the user study. This will focus on technical behaviour such as speech input/output, transcription, connection stability, interruption handling, and general system reliability. This testing is separate from the main user evaluation. Its purpose is to prepare the prototype so that the study can focus on participants’ experience of the implemented conversational design choices rather than on unresolved technical faults. 2
Experimental design Each participant will go through the following stages: Welcome, consent, and pre-task questionnaire – 10 minutes The researcher will welcome the participant, introduce the purpose of the study, provide the Participant Information Sheet, and obtain consent. Participants will then complete a short pre-task questionnaire covering only information directly relevant to the study. Participants will be told that the study evaluates the experience of interacting with the AI guide, not their knowledge of art, technology, or museums. They will also be informed not to reveal sensitive personal information during the interaction. Preparation and system familiarisation – 5 minutes The researcher will briefly explain how to start the web application and how to speak with the guide. The aim of this stage is not to train the participant in a complex interface, but simply to make them comfortable enough to begin the interaction. User interaction task – 25 minutes Participants will use Docent as a conversational guide in a simulated cultural setting (<i>see location</i>). Participants may follow a suggested route or tour, but will also be free to ask questions, follow their curiosity, and redirect the guide conversationally. The task is expected to last approximately 20–25 minutes, allowing enough time for the system to be experienced as a guide rather than a short demonstration. A possible participant instruction for this stage is: <i>“Use the AI guide to explore the exhibition. You may follow the suggested tour, ask questions whenever you like, interrupt or redirect the guide if it feels natural, and interact with it as you would with a human guide assisting you through the experience.”</i> Post-task questionnaire – 5 minutes After the interaction, participants will complete a post-task questionnaire about their experience. The questionnaire will focus on the perceived quality of the interaction, including naturalness, continuity, ease of spoken navigation, and overall suitability for a cultural setting. Because many features of conversation are experienced tacitly rather than consciously, the questionnaire will not be treated as the sole measure of interaction quality. Instead, it will be used alongside transcripts, observation notes, screen recordings (to observe instances that required touch input), and interview responses. Interview – 15 minutes A short semi-structured interview will be conducted after the questionnaire. The purpose of the interview is to allow participants to shed more light on their answers on the questionnaire. The interview will be audio-recorded for transcription. As with the interaction recordings, interview audio will be retained only as long as necessary to produce and verify an accurate transcript and will then be deleted. Data collection Participants will receive a Participant Information Sheet and will sign a Consent Form before taking part. They will be told that participation is voluntary, that they may stop during the study without 3	giving a reason, and that they should not disclose sensitive personal information during the interaction. The study will collect pre-task questionnaire responses, post-task questionnaire responses, researcher observation notes, conversation transcripts, interview transcripts, and, where appropriate, screen recordings. The main data source will be the transcript of the participant’s conversation with Docent, which will be used to examine how the interaction unfolded, including follow-up questions, interruptions, topic shifts, continuity, and moments of breakdown. Audio recordings of the interaction and interview may be temporarily retained only for transcription and transcript-checking. Because voice recordings may be identifiable, they will be treated as sensitive research data and deleted permanently within one week of participation, once the transcript has been checked. The remaining transcripts, notes, and questionnaire responses will be anonymised using participant identifiers such as P01, P02, and P03. Screen recordings may be used to identify moments where the participant needed to look at or touch the screen, restart an interaction, or perform an action that ideally would have been handled through speech. The study will not focus on technical logs such as latency, speech-processing timings, or model response timings; those issues will be addressed during developmental testing. Participants may request withdrawal of their data up to one week after participation. After this point, audio recordings will have been deleted and the remaining data will be anonymised, meaning it may no longer be possible to identify and remove an individual participant’s data. Anonymised or de-identified research data, such as transcripts, questionnaire responses, and researcher notes, will be retained for one year after completion of the study. The system is intended to use open-source or locally controlled components where possible. If any third-party speech-to-text, text-to-speech, transcription, or AI service is used, this will be disclosed to participants before the study. Bibliography Spiel, K., Haimson, O.L. and Lottridge, D. (2019) ‘How to do better with gender on surveys: A guide for HCI researchers’, <i>Interactions</i>, 26(4), pp. 62–65. doi: 10.1145/3338283. 4

II. Participant Information Sheet



Participant Information Sheet

Study title: Evaluating Design Considerations for Voice-Led Conversational AI Guides in Cultural Spaces

Version number and date: 1.0: 11.06.2026

Researcher's name: Olufemi Davies-Davies (Student / Researcher), Anna Xambó Sedo (Supervisor)

Queen Mary Research Ethics Committee reference number: To be allocated

Invitation paragraph

You are being invited to take part in a research study evaluating Docent, a conversational AI guide designed to deliver an art-gallery or cultural-tour experience through spoken interaction. Before deciding whether to take part, please read this information carefully. It explains the purpose of the study, what participation involves, what data will be collected, how that data will be used, and your rights as a participant.

What is the purpose of the study and what would taking part involve?

This study forms part of an MSc Artificial Intelligence thesis project. The project investigates how automated conversational systems can be designed to produce more natural, coherent, and guide-like interactions. The study is not testing your knowledge of art, museums, artificial intelligence, or technology. Instead, it is evaluating whether the design choices implemented in Docent support a good conversational experience. In particular, the study is

interested in how the system handles spoken interaction, follow-up questions, context, turn-taking, interruption, and continuation.

Why am I being invited?

You have been invited because you are an adult participant who may be able to take part in a spoken interaction with the system. You do not need any specialist knowledge of art, museums, AI, computer science, or conversational agents. The study is open to general adult users.

Do I have to take part?

No. Taking part is entirely voluntary. You may decide not to take part. You may also stop participating at any point during the study without giving a reason. You may request withdrawal of your data for up to one week after participation. After this point, audio recordings will have been deleted and the remaining transcripts and notes will be anonymised, meaning it may no longer be possible to identify and remove your individual data.

What are the possible benefits of taking part?

You may not receive a direct personal benefit from taking part. Your participation will help evaluate the design of conversational AI systems and may contribute to future recommendations for spoken AI guides and similar systems.

What are the possible disadvantages and risks of taking part?

The study involves minimal risk. The main risk is privacy-related, because spoken interaction may be temporarily recorded. To reduce this risk:

- recordings will only be used to produce and verify transcripts;

QM REC Participant Information Sheet template; Version 4.0 – 15 January 2026

- recordings will be deleted within one week;
- transcripts will be anonymised;
- you will be asked not to disclose sensitive personal information;
- directly identifiable demographic information will not be collected unless required.

There may also be minor inconvenience if the system fails, misunderstands you, or has to be restarted. If this happens, the researcher may pause or restart the session. You may stop taking part at any time.

Expenses and payments

Participants will not be compensated for taking part.

What information about me will you be collecting?

The study may collect the following data:

- your responses to the pre-task questionnaire;
- your responses to the post-task questionnaire;
- a transcript of your conversation with Docent;
- temporary audio recordings of the interaction, used only to check transcription accuracy;
- temporary audio recording of the interview, used only to produce and check a transcript;
- researcher observation notes;
- possible screen recording, used to identify moments where you had to look at or touch the screen;
- limited interaction information, such as the sequence of conversational turns or whether the system had to be restarted.

The main data source will be the conversation transcript

Your voice may be recorded temporarily during the interaction and interview. This is because the study needs accurate transcripts. Audio recordings may be used to check whether the transcript correctly represents what was said. Audio recordings will not be used in publications, presentations, or demonstrations. They will not be kept longer than necessary. Once the transcript has been checked and corrected, the audio recordings will be

QM REC Participant Information Sheet template; Version 4.0 – 15 January 2026

permanently deleted. Audio recordings will not be retained for longer than one week after your participation.

The study will avoid collecting directly identifiable personal data where possible. You will be assigned an anonymous participant code, such as P01, P02, or P03. Transcripts, notes, and questionnaire responses used in analysis will be stored under this code rather than under your name.

Your voice recording may be identifiable while it exists, which is why it will be treated carefully and deleted within one week. Anonymised or de-identified research data, such as transcripts, questionnaire responses, and researcher notes, will be retained securely for one year after completion of the study.

You should not share sensitive personal information during the interaction.

How will my data be stored and who will have access to it?

Your research data will be stored securely using Queen Mary University of London-approved storage, such as OneDrive, and will only be accessible to the student researcher and, where necessary, the project supervisor.

Your name and direct identifiers will not be used in the analysis. Instead, you will be assigned a participant code, such as P01, P02, or P03. Questionnaire responses, transcripts, researcher notes, and any screen-recording notes used in analysis will be stored under this participant code.

Audio recordings will be stored separately from anonymised research data and will only be kept temporarily for the purpose of producing and checking accurate transcripts. Once the transcript has been checked, the audio recordings will be permanently deleted. Audio recordings will not be shared with other participants, used in presentations, published, or included in the dissertation.

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Where possible, directly identifiable personal data will not be collected. If a signed consent form contains your name and signature, it will be stored separately from the research data. The link between your name and participant code will only be retained for as long as necessary to manage withdrawal requests and will not be used in the analysis.

When and how will my data be destroyed?

Temporary audio recordings of the interaction and interview will be permanently deleted within one week of participation, once the transcripts have been checked for accuracy.

After the audio recordings have been deleted, the remaining research data will consist of anonymised or de-identified material, such as questionnaire responses, conversation transcripts, interview transcripts, and researcher notes. These will be stored securely and may be retained for the period required for completion, assessment, and examination of the MSc dissertation, in line with Queen Mary University of London research data and records-retention requirements. Any signed consent forms will be stored separately from the research data and retained only as long as required for research governance and audit purposes. When data no longer needs to be retained, electronic files will be securely deleted.

How will my data be used and shared?

The information collected in this study will be used for analysis as part of an MSc Artificial Intelligence dissertation. The study findings may be used to discuss how design choices in conversational AI systems affect the quality of spoken interaction in cultural settings.

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The dissertation may include anonymised examples or short quotations from transcripts or interview responses. These will not include your name or information intended to identify you.

The data will not be made openly available in a public database. Audio recordings will not be published or shared. If anonymised findings are used in future academic outputs, such as presentations, reports, conference submissions, or journal articles, they will be presented in a way that does not identify individual participants.

Under what legal basis are you collecting this information?

- Queen Mary University of London **processes personal data** for research purposes in accordance with the lawful basis of 'public task'.
- Please read [Queen Mary's privacy notice for research participants](#) containing **important information about your personal data and your rights** in this respect.
- If you have any questions relating to data protection, please contact Queen Mary's Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

What will happen if I want to withdraw from this study?

You may stop taking part at any time during the study without giving a reason. If you withdraw during the session, the researcher will stop the study.

You may also request withdrawal of your data for up to one week after participation. If you make this request within that period, your audio recordings, transcripts, questionnaire responses, and notes will be destroyed where they can still be linked to you.

QM REC Participant Information Sheet template; Version 4.0 – 15 January 2026

After one week, audio recordings will have been deleted and the remaining research data will be anonymised or de-identified. At that stage, it may no longer be possible to identify which data belongs to you, so it may not be possible to remove your individual data from the study.

What should I do if I have concerns about this study?

- If you have any concerns about the manner in which the study was conducted, in the first instance, please **contact the researcher(s)** responsible for the study [Anna Xambó Sedo](#), [Olufemi Davies-Davies](#)
- If you have a complaint which you feel you cannot discuss with the researchers then you should **contact the Research Ethics Facilitators** by e-mail: research-ethics@qmul.ac.uk.
- When contacting the Research Ethics Facilitators, **please provide details** of the study title, description, the study and QM REC reference number (where possible), the researcher(s) involved, and details of the complaint you wish to make.

Who can I contact if I have any questions about this study?

Principal Investigator (or Supervisor for student projects)

[Anna Xambó Sedo](#)
a.xambosedo@qmul.ac.uk

Student Investigator (if applicable)

[Olufemi Davies-Davies](#)
itsolufemi@gmail.com
[+44 7849 636289](#)

QM REC Participant Information Sheet template; Version 4.0 – 15 January 2026

III. Consent Form



Consent Form

Title of Research Study: Evaluating Design Considerations for Voice-Led Conversational AI Guides in Cultural Spaces

Principal Investigator: Olufemi Davies-Davies (Student / Researcher), Anna Xambó Sedo (Supervisor)

Queen Mary Research Ethics Committee Ref: To be allocated

Thank you for your interest in this research.

Should you wish to participate in the study, **please consider the following statements.**

Before signing the consent form, you should **initial all or any of the statements that you agree with.**

Your signature **confirms that you are willing to participate in this research**, however you are reminded that **you are free to withdraw your participation** at any time.

QM REC Consent Form template (DOS-approved); Version 2.0: 15 January 2026

Statement	Please initial box
1. I confirm that I have read the Participant Information Sheet dated <u>11/06/2026</u> version <u>1.0</u> for the above study; or it has been read to me. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.	
2. I understand that my participation is voluntary and that I am free to stop taking part in the study at any time during participation without giving any reason and without my rights being affected.	
3. I understand that my data will be accessed by the student investigator, Olufemi Davies-Davies, and the project supervisor, Anna Xambó Sedo.	
4. I understand that my data will be securely stored using Queen Mary University of London-approved storage, such as OneDrive, and handled in accordance with Queen Mary University of London data protection guidelines. I understand that temporary audio recordings will be deleted within one week of participation, while anonymised or de-identified research data, such as transcripts, questionnaire responses, and researcher notes, will be retained for one year after completion of the study.	
5. I understand that I can access the information I have provided and request destruction of that information at any time within one week of taking part in the data collection.	

QM REC Consent Form template (DOS-approved); Version 2.0: 15 January 2026

I understand that after one week, I will not be able to request withdrawal of the information I have provided, because audio recordings will have been deleted and the remaining research data will be anonymised or de-identified.	
6. I agree to the interview and system interaction being audio-recorded for transcription and verification purposes.	
7. I understand that screen recordings may be used to analyze interaction patterns for research purposes only.	
8. I agree to take part in the above study.	

Participants should read [Queen Mary's privacy notice for research participants](#) which contains important information about your personal data and your rights in this respect.

If you have any **questions relating to data protection**, please contact Data Protection Officer, Queens' Building, Mile End Road, London, E1 4NS or data-protection@qmul.ac.uk or 020 7882 7596.

Participant ID Date Signature

QM REC Consent Form template (DOS-approved); Version 2.0: 15 January 2026

Name of person Date Signature
taking consent

I **Olufemi Davies-Davies** confirm that I have carefully explained the nature, demands and any foreseeable risks (where applicable) of the proposed research to the participant and provided a copy of this form.

Principal Investigator (or Supervisor for student projects)

Anna Xambó Sedo

a.xambosedo@qmul.ac.uk

Student Investigator (if applicable)

Olufemi Davies-Davies

itsolufemi@gmail.com

+44 7849 636289

QM REC Consent Form template (DOS-approved); Version 2.0: 15 January 2026

IV. Pre-task Questionnaires

Pre-task questionnaire

The pre-task questionnaire will run between 5 – 10 minutes. It will collect only information that is directly relevant to interpreting the spoken interaction. It will collect limited demographic and background information so that the study can describe the participant sample and interpret the spoken interaction. Participants will not be asked to provide their name, date of birth, address, student number, or other directly identifying information. All questions will include a “Prefer not to say” option where appropriate. The gender question is optional and allows participants to select more than one option. This follows inclusive HCI survey guidance (Spiel, Haimson and Lottridge, 2019) by avoiding a binary-only format and allowing participants to self-describe if they wish.

Possible questions include:

1. What is your age range?
 - ☐ 18–24
 - ☐ 25–34
 - ☐ 35–44
 - ☐ 45–54
 - ☐ 55–64
 - ☐ 65+
 - ☐ Prefer not to say
2. What is your gender? Please select all that apply
 - ☐ Woman
 - ☐ Man
 - ☐ Non-binary
 - ☐ Prefer to self-describe: _____
 - ☐ Prefer not to say
3. Is English your first language?
 - ☐ Yes
 - ☐ No
 - ☐ Prefer not to say
4. If English is not your first language, do you feel comfortable completing a spoken interaction in English for this study?
 - ☐ Yes
 - ☐ No

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- ☐ Prefer not to say
5. How often do you visit galleries, museums, exhibitions, or similar cultural spaces?
 - ☐ Never or almost never
 - ☐ A few times a year
 - ☐ About once a month
 - ☐ More than once a month
 - ☐ Prefer not to say
 6. How familiar are you with AI chatbots or conversational AI systems?
 - ☐ 1 = Not familiar at all
 - ☐ 7 = Very familiar
 7. How comfortable do you feel speaking aloud to a digital system?
 - ☐ 1 = Not comfortable at all
 - ☐ 7 = Very comfortable
 8. Do you have any privacy concerns about using a spoken AI system in this study?
 - ☐ Yes
 - ☐ No
 - ☐ Prefer not to say
 - ☐ If yes, please briefly describe: _____
 9. Do you understand that you should not share sensitive personal information during the interaction?
 - ☐ Yes
 - ☐ No

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V. Post-task Questionnaires

Appendix B – Post-task questionnaire

The post-task questionnaire will focus on the participant's felt experience of the interaction. It will not be treated as a complete measure of conversational quality, since many conversational features are tacit and may not be easy for participants to describe directly.

Participants will be asked to rate the following statements on a 1–7 scale, where 1 means “strongly disagree” and 7 means “strongly agree”.

Section A — Overall experience

1. The interaction felt natural.
2. I felt comfortable speaking with the guide.
3. The guide was easy to follow.
4. I would be willing to use a system like this in a real cultural setting.
5. The experience felt more like being guided than using a conventional app.

Section B — Conversation and continuity

6. The guide responded appropriately to what I said.
7. The guide seemed to maintain the thread of the conversation.
8. I was able to ask follow-up questions naturally.
9. The guide handled changes in topic or direction well.

Section C — Spoken navigation

11. I was able to use speech as the main way of interacting with the guide.
13. When I wanted the guide to do something, I felt able to ask for it in ordinary language.

Section D — Breakdowns and limitations

15. There were moments where the guide misunderstood me.
16. There were moments where the guide's response felt unnatural.
17. There were moments where I had to change how I spoke in order to get the system to respond properly.
18. There were moments where the experience felt interrupted by the technology.

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Appendix C — Post-questionnaire clarification interview

After completing the post-task questionnaire, participants will take part in a short semi-structured clarification interview. The purpose of the interview is not to repeat the questionnaire, but to give participants the opportunity to explain their answers in more detail and refer to specific moments from the interaction.

The interview is expected to last approximately 5–10 minutes. Only a subset of the questions below will be used with each participant. The researcher will select the most relevant questions based on the participant's questionnaire responses and what occurred during the session.

1. Looking at your questionnaire responses, are there any answers you would like to explain in more detail?
2. Were there any particular moments in the interaction that influenced your ratings?
3. You rated the interaction on naturalness and comfort. What made the interaction feel natural or unnatural to you?
4. You rated the guide on continuity and follow-up conversation. Were there moments where it seemed to maintain or lose the thread of the conversation?
5. You rated the experience of using speech as the main way of interacting. Did speech feel like an appropriate way to use the guide? Why or why not?
6. Were there any moments where you felt you had to change how you spoke in order for the system to understand or respond properly?
7. Were there any moments where the technology interrupted or distracted from the experience?
8. Was there anything in your questionnaire responses that you found difficult to answer, or that you think needs more explanation?
9. Based on your experience, what would be the most important improvement to the guide?
10. Is there anything else about the interaction that you think is important for the study to know?

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VI. Ethics Panel Approval



School of Electronic Engineering
& Computer Science
Prater Landing Building, 10 Godward Square
London E1 4FZ

Devolved School Research Ethics Committee
Email: eeecs-ethics@qmul.ac.uk

22 June 2026

Dr Anna Xambó Sedó
School of Electronic Engineering and Computer Sciences
Queen Mary University of London

Cc: Olufemi Davies-Davies

Dear Anna Xambó Sedó

RE: DSEEC26.072 - Evaluating Design Considerations for Voice-Led Conversational AI Guides in Cultural Spaces.

Having received a Research Ethics Application for the study detailed above, the proposed work has been deemed low-risk and will not require the scrutiny of the full Queen Mary Research Ethics Committee. It has been reviewed by the Electronic Engineering and Computer Science Devolved School Research Ethics Committee (EECS DSREC) and I am pleased to inform you that, based on the information provided, this application has now been **approved**.

- Date of Approval: 22 June 2026
- Approval End Date: 21 June 2029

Extensions and Amendments:

If you propose to make an [amendment to the research after approval](#) or extend the study beyond the date of the approval expiry date given above, then the EECS DSREC should be contacted (eeecs-ethics@qmul.ac.uk).

Adverse Events:

In case of any unexpected or adverse events or deviations from the proposed protocol as per the original application, please contact the EECS DSREC at eeecs-ethics@qmul.ac.uk within 5 days of the event.

QM REC Approval letter – Low-risk studies
V5.0

Health and safety:

Researchers should exercise extra care and precautions when conducting research at this time. There are ethical issues that are of heightened importance to consider when conducting research with human participants.

Researchers are responsible for checking the local and national governmental advice and for being aware of any necessary restrictions that may impact their study. In particular, research that involves overseas travel or face-to-face research must regularly check for updates and follow the latest guidance on Health and Safety.

We wish you the best for your research.

Yours faithfully,

Alvaro Bort

Alvaro Bort
On behalf of the Electronic Engineering and Computer Science DSREC

XIII. APPENDIX B

ECS750P/ECS753P/ECS754P– MSc Project Module

Generative AI Tools – Student Accountability Statement

Student Name: Olufemi Davies-Davies

Project Title: Docent: Design Considerations for Voice-Led Conversational AI in Cultural Spaces

Supervisor Name: Anna Xambo Sedo

1. Have you used Generative AI tools for your project?

YES ✓

NO

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	5.6	OpenAI	https://chat.openai.com/
Gemini notebook	2.0	Google	https://notebook.google.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Literature Review	Used Generative AI to investigate academic papers	Notebook LM is a great tool for probing papers, as it can deliver the content of a paper as a “podcast” in chart or videos explaining the concept, as well as answering questions on said paper
Software development	Employed generative AI for error location, general debugging, planning development and visualizing data flow	Gen AI tools are very fast in locating and identifying errors even before runtime. Furthermore, in the core of deliberating new programs to implement, or new tools to employ, gen AI is an effective partner in deliberating a plan to most efficient implement a new feature.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: _____

A handwritten signature in black ink, appearing to be 'F. Davies', written over a horizontal line.

Date: _____19/08/2026_____